

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the Smart Lender codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S. No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP8 followed across all Python scripts
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for data, model, templates, and static files
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. loan_amount, credit_history, applicant_income
4	Function / Method Names	Functions are descriptively named	Yes	e.g. preprocess_data(), train_model(), predict_loan_status()
5	Code Comments	Inline and block comments explain logic	Partial	Core logic commented; minor utility functions need more comments
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for preprocessing, training, and Flask app
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused experimental model code

S. No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
				removed before deployment
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Try-except added for form input validation; needs broader coverage

Reusable Components / Modules:

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	preprocess_data()	Python (Pandas, NumPy)	Used in both training pipeline and Flask prediction endpoint	High
2	model_utils.py (load/save model)	Python (Pickle/Joblib)	Used across training script and Flask app for model persistence	High
3	validate_input()	Python (Flask)	Used in web form submission and batch upload feature	Medium
4	base.html template	HTML / Jinja2	Used as base layout for all web application pages	High
5	evaluate_model()	Python (Scikit-learn, XGBoost)	Used to evaluate Decision Tree, Random Forest, KNN, and XGBoost models	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Well-organized into preprocessing, model, and web app modules
Readability	4	Clear naming conventions and logical flow throughout
Reusability	4	Core functions reused across training and prediction pipelines
Documentation / Comments	3	Adequate comments; could be expanded for utility functions
Overall Score	4	Solid, maintainable codebase suitable for future extension